

Task 4: Aggregate Functions & Grouping

Comprehensive SQL Queries & Technical Interview Preparation

SUBMITTED BY

G. Pavana Deepika

TOOLS USED

SQLite / MySQL Workbench

KEY CONCEPTS

Aggregation, Grouping,
Having

Part 1: SQL Query Implementations

1. Total & Average Salary by Department

```
SELECT department_id,  
       COUNT(employee_id) AS total_employees,  
       SUM(salary) AS total_salary,  
       ROUND(AVG(salary), 2) AS average_salary  
FROM employees  
GROUP BY department_id;
```

2. Filter Groups with Average Salary > 50,000

```
SELECT department_id,  
       ROUND(AVG(salary), 2) AS average_salary  
FROM employees  
GROUP BY department_id  
HAVING AVG(salary) > 50000;
```

3. Find Highest Salary per Department

```
SELECT department_id,  
       MAX(salary) AS highest_salary  
FROM employees  
GROUP BY department_id;
```

4. Count Unique Job Roles Across Company

```
SELECT COUNT(DISTINCT job_title) AS unique_job_roles  
FROM employees;
```

Part 2: Technical Interview Questions & Answers

1. What is GROUP BY?

GROUP BY collates rows sharing identical values across specified columns into summary rows (e.g., grouping employees by department to perform aggregate computations).

2. Difference between WHERE and HAVING?

WHERE filters individual rows *before* any aggregate calculations or grouping occur. HAVING filters aggregated groups *after* the GROUP BY clause is applied.

3. How does COUNT(*) differ from COUNT(column)?

COUNT (*) counts every row in the target table/group including NULL values, whereas COUNT (column) counts only non-null entries in that specific column.

4. Can you group by multiple columns?

Yes (e.g., GROUP BY department_id, job_title). This creates unique groups for each distinct combination of values across all specified columns.

5. What is ROUND() used for?

ROUND () limits numerical or floating-point values to a specified number of decimal places for cleaner reporting (e.g., ROUND(AVG(salary), 2)).

6. How do you find the highest salary by department?

By combining the MAX () function with a GROUP BY clause:

```
SELECT department_id, MAX(salary) FROM employees GROUP BY department_id;
```

7. What is the default behavior of GROUP BY?

It collapses rows with identical key values into a single summary row per unique combination. In many relational engines, output groups are also implicitly organized by those keys.

8. Explain AVG and SUM.

SUM () computes the total mathematical sum of a numeric column, while AVG () calculates the arithmetic mean of a column, automatically ignoring NULL values.

9. How to count distinct values?

By nesting the DISTINCT keyword inside the COUNT () aggregate function, such as COUNT (DISTINCT column_name).

10. What is an aggregate function?

An aggregate function takes multiple input values across multiple rows and processes them into a single summary output value (e.g., SUM, COUNT, AVG, MIN, MAX).